UNIVERSAL COLLEGE OF ENGINEERING

Industrial Internship Report on

”Prediction of agriculture production”

Prepared by

[Priyanka Das]

Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks’ time. My project was (Tell about ur Project) This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

TABLE OF CONTENTS

1 Preface 3
2 Introduction 5
2.1 About UniConverge Technologies Pvt Ltd 5
2.2 About upskill Campus 9
2.3 Objective 11
2.4 Reference 11
2.5 Glossary 11
3 Problem Statement 12
4 Existing and Proposed solution 13
5 Proposed Design/ Model 14
5.1 High Level Diagram (if applicable) 14
5.2 Low Level Diagram (if applicable) 15
5.3 Interfaces (if applicable) 16
6 Performance Test 18
6.1 Test Plan/ Test Cases 18
6.2 Test Procedure 18
6.3 Performance Outcome 19
7 My learnings 20
8 Future work scope 21

1 Preface

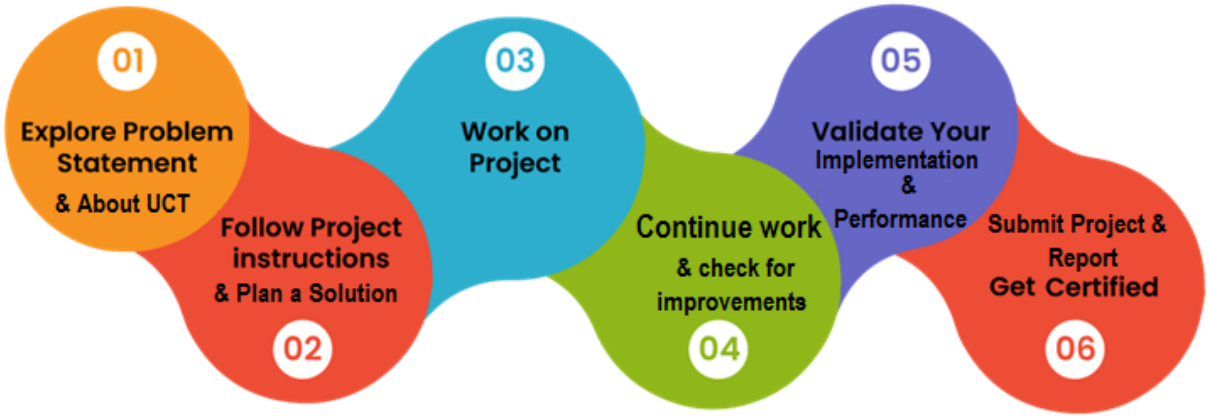
Summary of the whole 6 weeks' work.

About need of relevant Internship in career development.

Brief about Your project/problem statement.

Opportunity given by USC/UCT.

How Program was planned



Your Learnings and overall experience.

During this internship, I gained valuable practical knowledge and hands-on experience in applying theoretical concepts to real-world problems. I improved my understanding of Python programming, data preprocessing, and machine learning techniques such as regression models. I also learned how to work with datasets, handle errors, and build a complete project from scratch, including frontend and backend integration.

This experience enhanced my problem-solving skills, logical thinking, and ability to work independently as well as in a structured environment. I also developed skills in using tools like Flask for web development and GitHub for version control. Overall, the internship was a great learning experience that helped me grow both technically and professionally.

Thank to all (USC / UCT Team), who have helped you directly or indirectly.

I would like to express my sincere gratitude to all those who supported and guided me throughout this internship.

I am especially thankful to my mentors and faculty members for their valuable guidance, continuous support, and encouragement during the project. I would also like to thank the USC/UCT team for providing this opportunity and structured learning environment.

I am grateful to my friends and peers who helped me by sharing ideas, resolving doubts, and motivating me throughout the internship. Lastly, I would like to thank my family for their constant support and encouragement.

Your message to your juniors and peers.

I would like to encourage my juniors and peers to actively participate in internships and practical projects, as they provide real-world exposure beyond textbooks. Consistent practice, curiosity, and willingness to learn are key to success in this field.

Do not hesitate to explore new technologies, ask questions, and learn from mistakes. Building projects and gaining hands-on experience will greatly help in improving your skills and confidence. Make the most of every opportunity and keep learning continuously.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

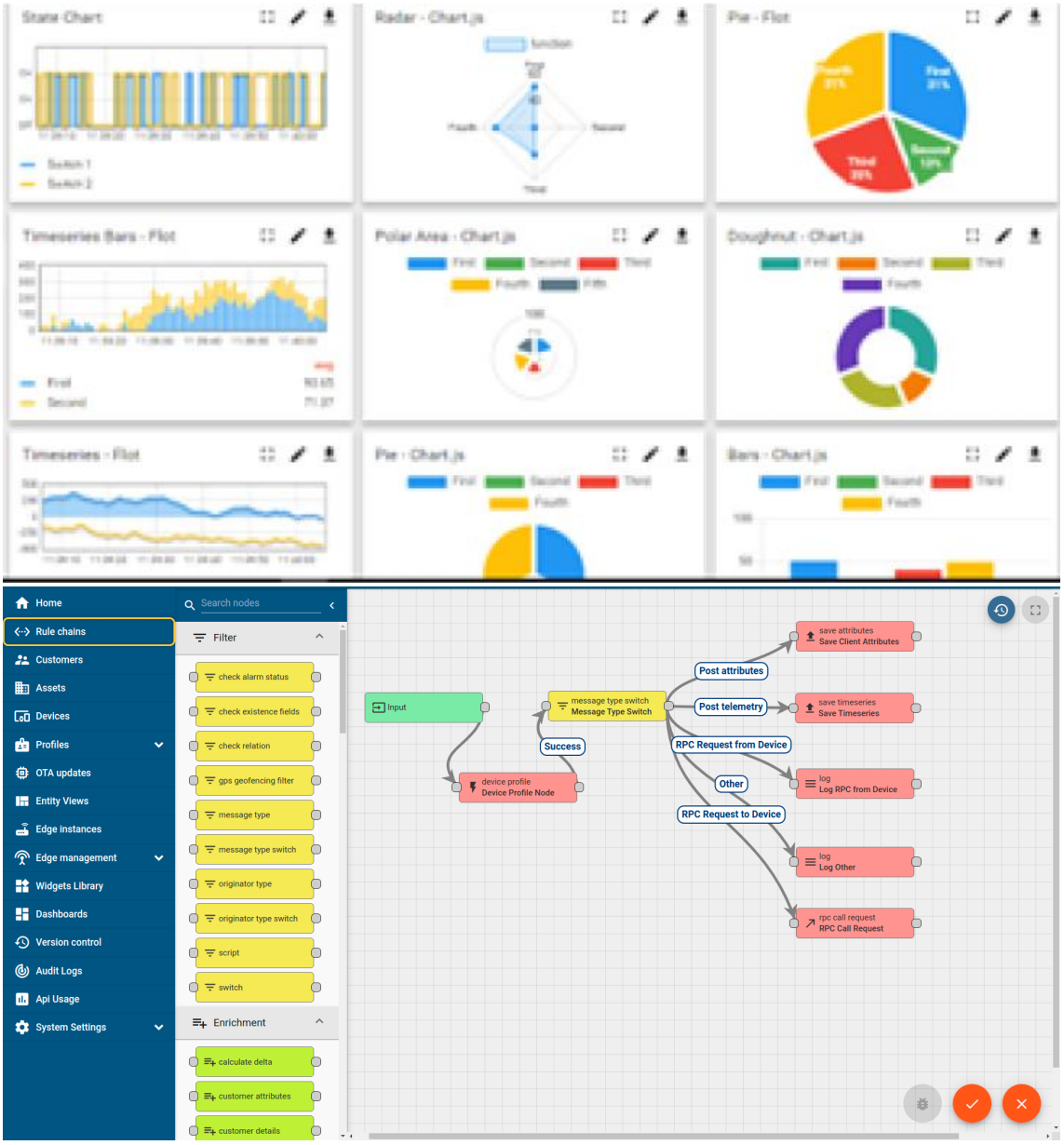
i. UCT IoT Platform (uct Insight)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

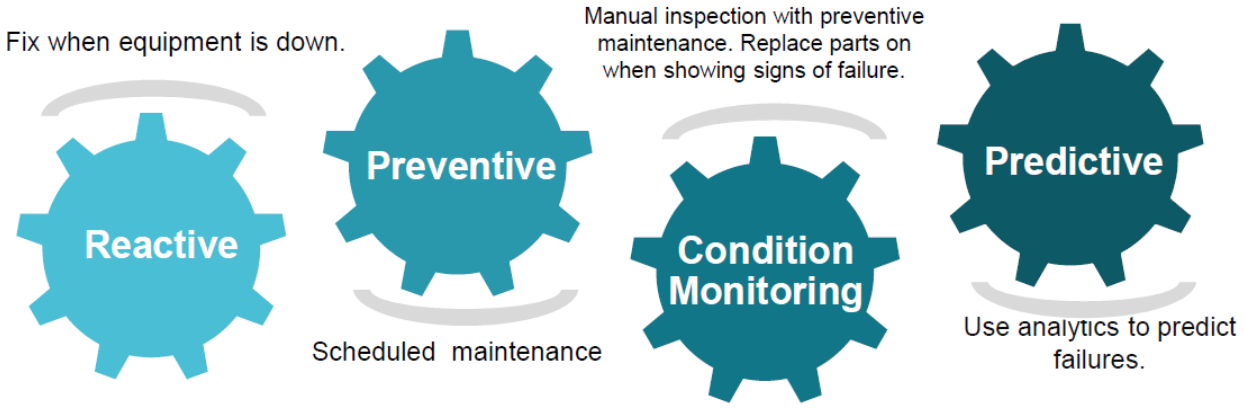
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

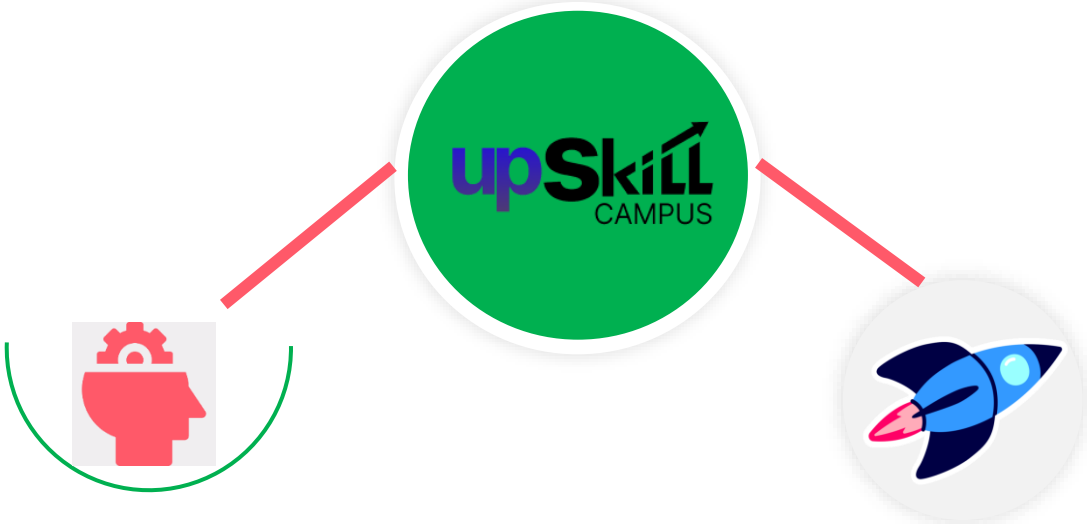
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

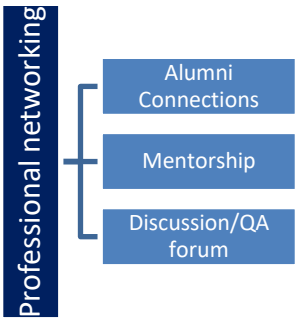
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.

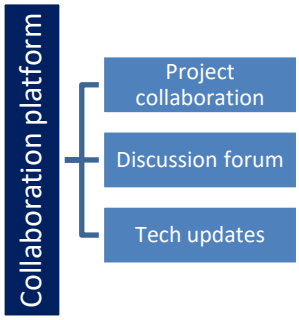


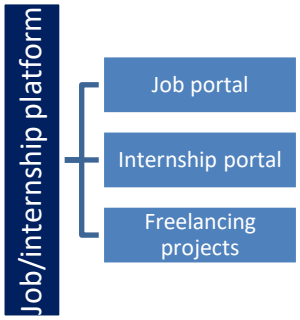
Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/







2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- ☛ get practical experience of working in the industry.
- ☛ to solve real world problems.
- ☛ to have improved job prospects.
- ☛ to have Improved understanding of our field and its applications.
- ☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1]Government of India Open Data Portal – Agricultural Dataset https://data.gov.in/

[2] Documentation of Python Programming Language https://docs.python.org/3/

[3]Scikit-learn Machine Learning Library Documentation https://scikit-learn.org/

2.6 Glossary

Terms	Acronym
Machine Learning	ML
Artificial Intelligence	AI
Model	Trained algorithm used for prediction
Prediction	Output generated by the model
Dataset	Collection of data used for training/testing

3 Problem Statement

In the assigned problem statement

Agriculture is a vital sector in India, supporting the livelihood of a large portion of the population. However, crop production is influenced by multiple factors such as crop type, variety, seasonal conditions, cultivation cost, and geographical location. Due to these varying factors, farmers often face uncertainty in predicting the expected production, which can lead to financial losses and inefficient resource utilization.

The problem addressed in this project is to analyze historical agricultural data (2001–2014) and develop a predictive model to estimate crop production based on relevant features such as crop, state, season, quantity, and cost. By leveraging data analysis and machine learning techniques, the project aims to provide insights that can support better decision-making in agriculture.

This system will help in:

- Understanding patterns in crop production
- Predicting future production trends
- Assisting farmers and policymakers in planning and resource management

4 Existing and Proposed solution

Provide summary of existing solutions provided by others, what are their limitations?

- Existing Solutions

Traditional statistical methods like regression and time-series are used for crop prediction.
Machine learning models such as Linear Regression and Random Forest improve prediction accuracy.
Government systems provide agricultural forecasts based on large datasets.
Some web/mobile apps offer crop prediction and farming support.

- Limitations

Limited factors considered (lack of real-time data like weather and soil).
Accuracy issues with small or incomplete datasets.
Complex models are not user-friendly for farmers.
Dependence on historical data only.

Some systems are costly and not easily accessible.What is your proposed solution?

To overcome the limitations of costly and less accessible systems, this project proposes a simple, low-cost, and user-friendly web-based application for predicting crop production.
Uses machine learning (Linear Regression / Random Forest) on historical agricultural data.
Provides an easy interface where users can input crop, state, and cost details.
Delivers quick predictions without requiring technical knowledge.
Can be accessed through a browser, making it widely available.
Designed to work with minimal resources, making it affordable and scalable.

What value addition are you planning?

This project adds value by providing a simple, accessible, and data-driven solution for crop production prediction.
Introduces a user-friendly web interface for easy input and quick results.
Uses machine learning models to improve prediction accuracy compared to basic methods.
Helps in better decision-making for crop selection and resource planning.
Reduces dependency on complex and costly systems.
Provides visual insights and analysis for better understanding of agricultural trends.

4.1 Code submission (Github link): https://github.com/priyankadas-star/upskillcampus

4.2 Report submission (Github link) : https://github.com/priyankadas-star/upskillcampus

5 Proposed Design/ Model

Given more details about design flow of your solution. This is applicable for all domains. DS/ML Students can cover it after they have their algorithm implementation. There is always a start, intermediate stages and then final outcome.

5.1.1 High Level Diagram (if applicable)

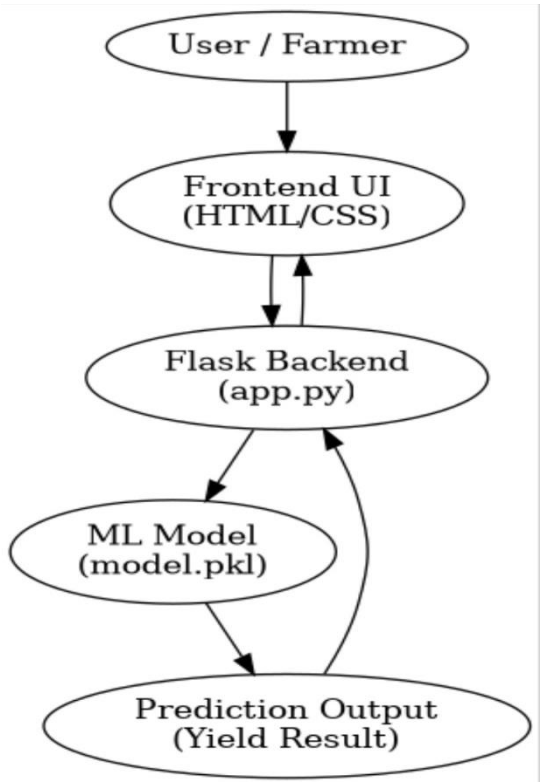
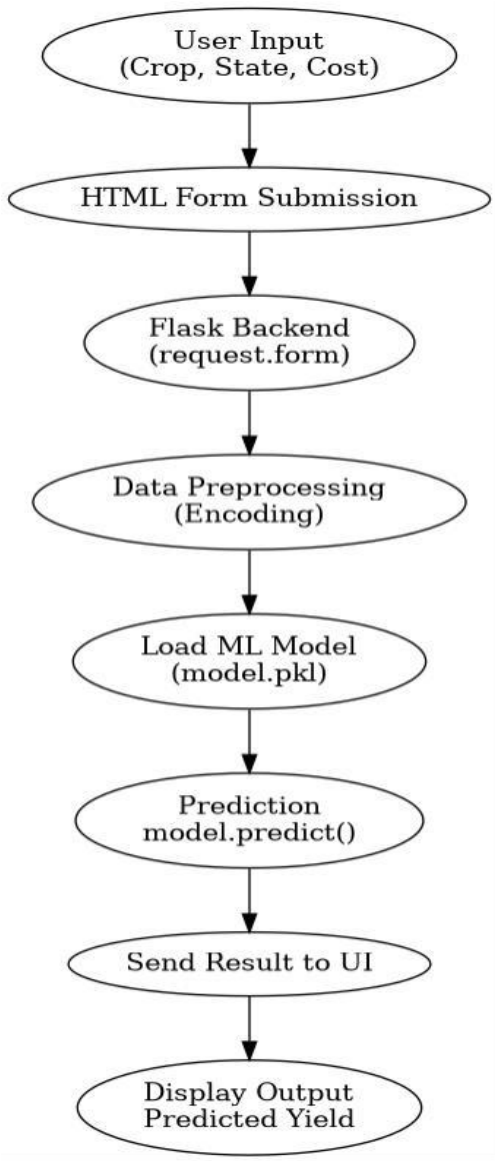


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM



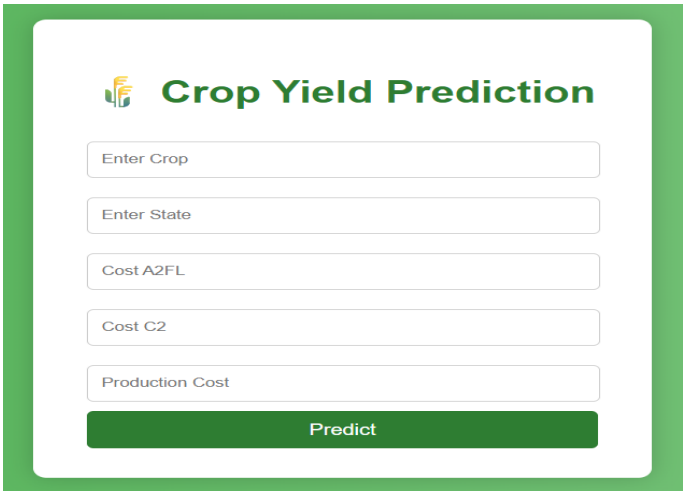
5.2 Low Level Diagram (if applicable)

5.3 Interfaces (if applicable)

Update with Block Diagrams, Data flow, protocols, FLOW Charts, State Machines, Memory Buffer Management.

1. User Interface (UI)

- A web-based interface developed using HTML, CSS.
- Allows user to input:
  - Crop
  -
  - State
  - Cost values
- Displays predicted output (Yield/Production).

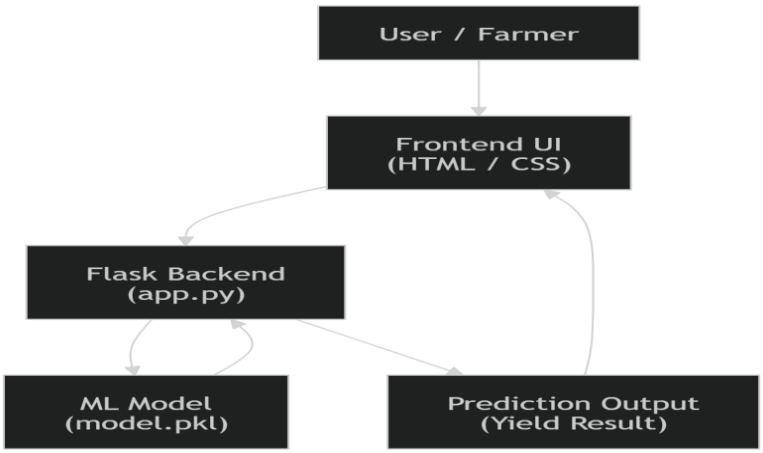


2. System Interface (Block Diagram)

The system consists of three main components:

- Frontend (User Interface)
- Backend (Flask Server)
- Machine Learning Model

The frontend sends user input to backend, backend processes data and calls the ML model, and the result is returned to the user.



3. Data Flow Diagram (DFD)

- User → Input Data
- Backend → Processing
- Model → Prediction
- Output → User

Write 2–3 lines explanation:

Data flows from user input through the backend where preprocessing and prediction occur, and the result is sent back to the interface.

4. Flowchart

1. Start
2. Enter Input
3. Validate Input
4. Encode Data
5. Load Model
6. Predict Output
7. Display Result
8. End

5. Protocols Used

Keep it simple:

- **HTTP Protocol** → Communication between frontend and backend
- **REST API (Flask)** → Handles requests and responses
  The system uses HTTP protocol for communication and Flask-based APIs to process user requests.

6. Memory / Buffer Management (Simple Explanation)

You don't need deep OS-level explanation. Write:

- Data is temporarily stored in memory during processing
- Input is passed to the model using NumPy arrays
- No permanent storage is required

6 Performance Test

This is very important part and defines why this work is meant of Real industries, instead of being just academic project.

Here we need to first find the constraints.

How those constraints were taken care in your design?

What were test results around those constraints?

Constraints can be e.g. memory, MIPS (speed, operations per second), accuracy, durability, power consumption etc.

In case you could not test them, but still you should mention how identified constraints can impact your design, and what are recommendations to handle them.

6.1 Test Plan/ Test Cases

The testing of the system is performed to ensure that the application works correctly and provides accurate predictions. The system is tested for functionality, input validation, and output correctness.

7 Objectives of Testing

- To verify correct prediction of crop yield
- To check proper handling of user inputs
- To ensure smooth interaction between frontend and backend
- To identify and fix errors in the system

7.1 Test Procedure

The testing of the system is carried out in a systematic manner to ensure that all components of the application function correctly and efficiently. The following steps were followed during testing:

7.2 Step-by-Step Procedure

1. **Start the Application**
   - Run the Flask application using `python app.py`.
   - Ensure the server starts without errors.
2. **Open the Web Interface**
   - Open a web browser and navigate to `http://127.0.0.1:5000/`.
   - Verify that the user interface loads properly.
3. **Enter Input Data**
   - Provide valid inputs such as crop name, state, and cost values in the form fields.
4. **Submit the Form**
   - Click on the **Predict** button to send data to the backend.

5. **Backend Processing**
   - The Flask server receives the input.
   - Data is preprocessed (encoding of crop and state).
   - The trained machine learning model is loaded.
6. **Prediction Generation**
   - The model processes the input data and generates the predicted output.
7. **Display Output**
   - The predicted yield is displayed on the web page.
8. **Validation of Results**
   - Compare the output with expected results for correctness.
9. **Error Handling Check**
   - Test with invalid or empty inputs to verify proper error handling.

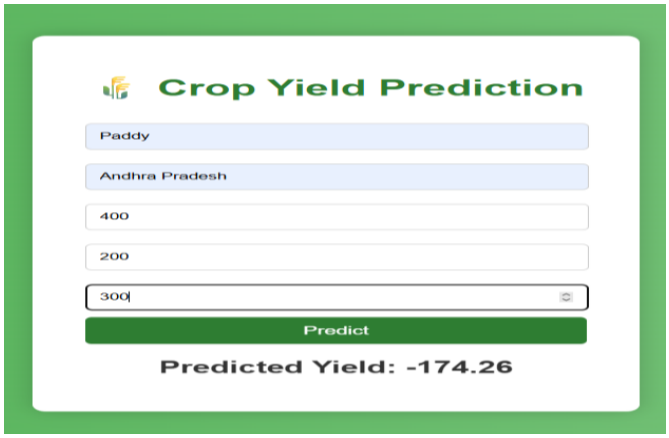
7.3 Performance Outcome

The performance of the developed system was evaluated based on accuracy, response time, and usability. The machine learning model was able to generate predictions with acceptable accuracy using the available dataset.

The system provides **quick responses**, as the prediction is generated instantly after submitting the input. The integration between the frontend and backend works smoothly, ensuring efficient data flow and minimal delay.

The application is **user-friendly**, allowing users to easily input data and obtain results without requiring technical knowledge. The model performs well for valid inputs, although the accuracy may be limited due to the small size of the dataset and lack of real-time parameters.

Overall, the system demonstrates satisfactory performance and successfully meets the objectives of predicting agricultural crop production.



8 My learnings

You should provide summary of your overall learning and how it would help you in your career growth.

During this internship, I gained valuable knowledge and practical experience in applying theoretical concepts to real-world problems. I developed a strong understanding of Python programming, data preprocessing, and machine learning techniques used for prediction. I learned how to work with datasets, clean and transform data, and build models to generate meaningful results.

I also gained hands-on experience in developing a complete project, including integrating the machine learning model with a web application using Flask. Additionally, I improved my skills in debugging, problem-solving, and using tools like GitHub for version control.

This internship enhanced my ability to think logically, work independently, and manage tasks efficiently. The experience will help me in my career growth by strengthening my technical skills, improving my confidence, and preparing me for real-world industry challenges in the field of data analysis and software development.

9 Future work scope

You can put some ideas that you could not work due to time limitation but can be taken in future.

Although the project successfully predicts agricultural crop production, there are several improvements that can be implemented in the future.

- Integration of real-time data such as weather conditions, rainfall, and soil quality to improve prediction accuracy.
- Use of advanced machine learning models like Random Forest and XGBoost for better performance.
- Development of a mobile application for easier accessibility by farmers.
- Addition of crop recommendation system based on region and season.
- Enhancement of the user interface with dropdown options and better visualization tools.
- Deployment of the system on a cloud platform to make it accessible online.
- Expansion of dataset to include more years and regions for improved reliability.